Quick Work

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M T W T F S S

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Assignment 4

1 Resolution is the smallest change is analog input that is ADC can detect

Formula

$$\text{Resolution} = \frac{V_{ref}}{2^n}$$

$$= \frac{5}{2^{10}} = \frac{5}{1024} \approx 0.00488\,V$$

converting to mv : $0.00488 \sim 4.88\,mV$

2. 6 bit ADC ($V_{ref} = 10V$, $V_{in} = 3.7V$)

$$\frac{10}{2^6} = \frac{10}{64} = 0.15625\,V$$

Digital code

$$\Delta = \frac{V_{in}}{\text{Resolution}} = \frac{3.7}{0.15625} = 23.68$$

Since Output is always integer

$\approx 24 = (011000)_2$

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Q3 8-bit DAC with full-scale 12.75 V

$$V_{out} = \frac{D \times V_{FS}}{2^{n}-1}$$

$10110010_2 = 178$

Step 2 calculate step size

$$\frac{12.75}{255} = 0.05\,V$$

$V_{out} = 178 \times 0.005 \times 8.9\,V$

Q4 $01101_2 = 13$

$V_{out} = D \times k = 13 \times 0.5 = 6.5\,V$

output voltage = 6.5 V

Q5 $D = \frac{V_{in}}{\text{Resolution}} = \frac{9.3}{1} = 9.3$

$D = 9$

$9 = (1001)_2$

Digital output = 1001